

A Late-Time Background Deformation of Λ CDM: Full-Likelihood Constraints on the ECTI-P Model

Felipe D. J. M. Patellis

2026

Abstract

We investigate a minimal phenomenological extension of Λ CDM, denoted ECTI-P, which introduces a localized late-time deformation of the effective dark-energy density while preserving the standard early-Universe evolution. The model is implemented at the background level and tested using a full-likelihood pipeline combining Planck 2018 TT/TE/EE data with late-time probes including Pantheon+ supernovae, DESI BAO, redshift-space distortions, and weak lensing. Under a strictly controlled comparison with identical datasets, likelihoods, and parameter space, ECTI-P yields an improvement of $\Delta\chi^2 = -15.69$ at the maximum a posteriori. The improvement is driven primarily by Type Ia supernovae, with additional contribution from KiDS, while the CMB likelihood remains essentially preserved and BAO constraints show a moderate degradation. The model shifts the MAP solution toward a higher $H_0 = 70.91 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and a lower $S_8 = 0.8117$, compared to $H_0 = 67.94 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $S_8 = 0.8285$ in Λ CDM. We emphasize that ECTI-P is not proposed as a fundamental theory, but as a phenomenological framework to isolate potential late-time deviations from Λ CDM.

1 Introduction

The Λ CDM model [1] provides an exceptionally successful description of the large-scale Universe, accurately reproducing observations of the cosmic microwave background (CMB) [2], baryon acoustic oscillations (BAO), and large-scale structure. However, persistent discrepancies in late-time probes, notably in the Hubble constant H_0 and the clustering amplitude S_8 , continue to motivate the exploration of extensions beyond a strictly constant dark-energy component.

A key feature of these discrepancies is their apparent localization in the late-time Universe, suggesting that modifications confined to low redshift may be sufficient to improve the global fit without altering early-Universe physics. In this work, we investigate such a minimal extension, denoted ECTI-P, which introduces a localized deformation of the effective dark-energy density that becomes relevant only at late times and is suppressed at high redshift.

2 Model Definition

The ECTI-P background expansion is defined through the effective dark-energy density:

$$\frac{\rho_{\text{DE}}^{\text{ECTI}}(z)}{\rho_{\text{DE}}(0)} = \exp[\beta(1 - \exp(-z/z_t))]. \quad (1)$$

The corresponding expansion history is:

$$E^2(z) \equiv \frac{H^2(z)}{H_0^2} = \Omega_m(1+z)^3 + (1 - \Omega_m) \exp[\beta(1 - \exp(-z/z_t))]. \quad (2)$$

For $\beta = 0$, ECTI-P reduces exactly to Λ CDM.

These values were calibrated in an independent background-only analysis using late-time datasets (Pantheon+, DESI BAO, RSD, KiDS-1000), documented at https://github.com/f-patellis/ECTI-P_vs_LCDM. Fixing them reduces the look-elsewhere effect and ensures that the present work constitutes a consistency test rather than a parameter-fitting exercise. The deformation is exponentially suppressed at high redshift, ensuring that early-Universe observables such as the CMB remain unaffected.

3 Methodology

3.1 Implementation

The model is implemented within the CLASS Boltzmann solver [8], with sampling performed using Cobaya [9]. A hybrid approach is adopted:

- Background: modified (ECTI-P)
- Perturbations: standard Λ CDM

The full pipeline and MCMC chains used in this analysis are publicly available (see Data and Code Availability section).

3.2 Datasets

The analysis combines:

- Planck 2018 TT/TE/EE [2]
- Pantheon+ SN Ia [3]
- DESI 2024 BAO [4]
- Redshift-space distortions (RSD, $f\sigma_8$) from BOSS and eBOSS [5, 6]
- KiDS-1000 weak lensing [7]

The parameter space is identical to Λ CDM, with no additional free parameters introduced in the likelihood exploration.

4 Results

4.1 Global fit

At the maximum a posteriori, the best-fit values are:

$$\chi^2_{\Lambda\text{CDM}} = 4677.72, \quad \chi^2_{\text{ECTI-P}} = 4662.03, \quad (3)$$

corresponding to

$$\Delta\chi^2 = \chi^2_{\text{ECTI-P}} - \chi^2_{\Lambda\text{CDM}} = -15.69. \quad (4)$$

4.2 Posterior shifts

ECTI-P induces the following MAP-level parameter shifts:

- H_0 : $67.94 \rightarrow 70.91 \text{ km s}^{-1} \text{ Mpc}^{-1}$
- Ω_m : $0.3082 \rightarrow 0.2815$
- σ_8 : $0.8174 \rightarrow 0.8380$
- S_8 : $0.8285 \rightarrow 0.8117$

These shifts are illustrated in the full posterior comparison shown in Fig. 1, where ECTI-P produces a coherent displacement of the parameter space relative to Λ CDM.

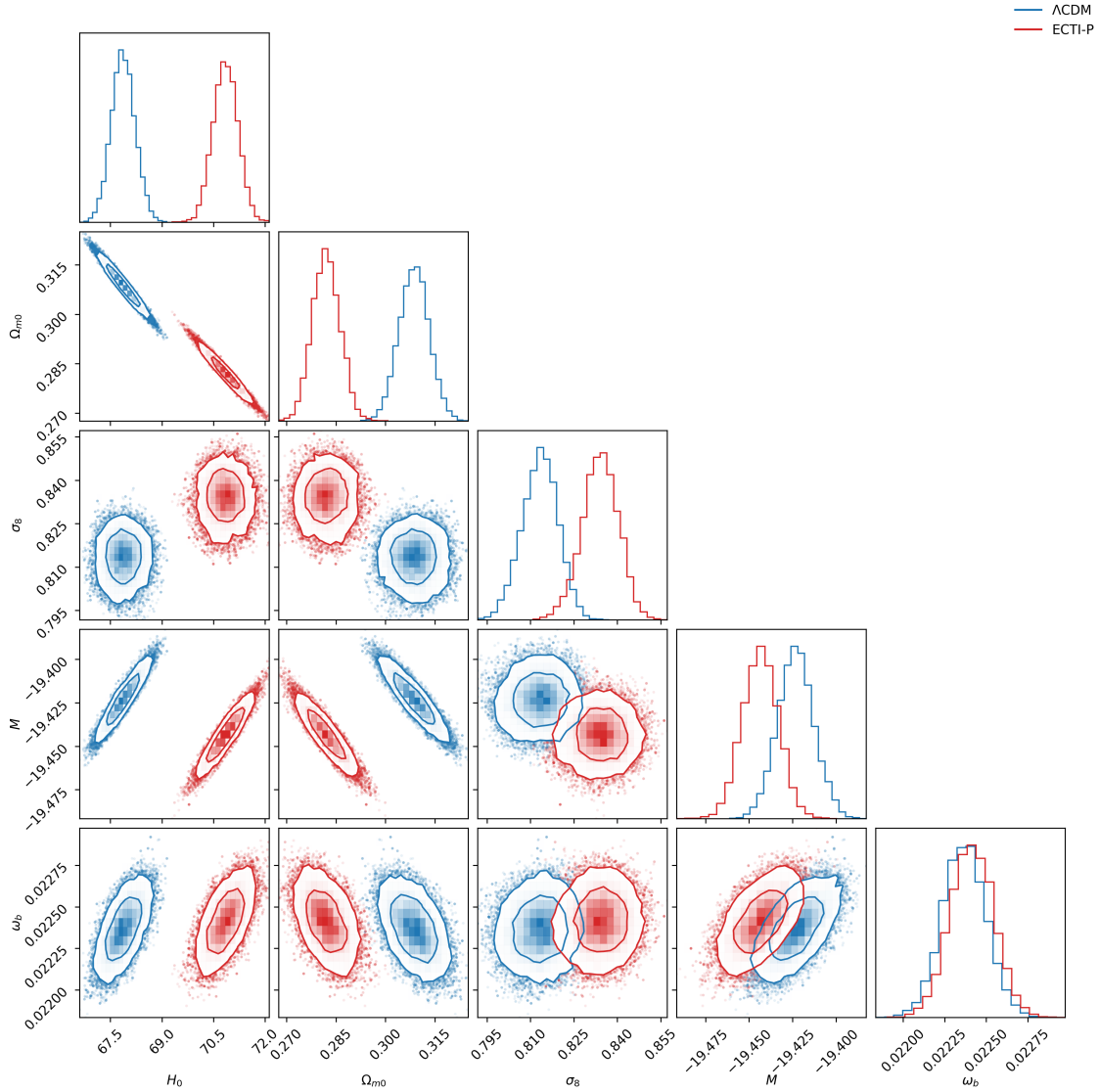


Figure 1: Posterior comparison between Λ CDM (blue) and ECTI-P (red).

In particular, the model shifts the solution toward higher H_0 and lower S_8 , while maintaining well-constrained posteriors.

The corresponding MAP parameter values are summarized in Fig. 2.

MAP summary table

	chi2	H0	Omega_m	sigma8	S8	Delta chi2
LCDM	4677.72	67.94	0.3082	0.8174	0.8285	0.00
ECTI-P	4662.03	70.91	0.2815	0.8380	0.8117	-15.69

Figure 2: MAP parameter summary for Λ CDM and ECTI-P.

This behavior is further illustrated in the H_0 – S_8 plane (Fig. 3), where ECTI-P moves the solution in the direction favored by late-time probes.

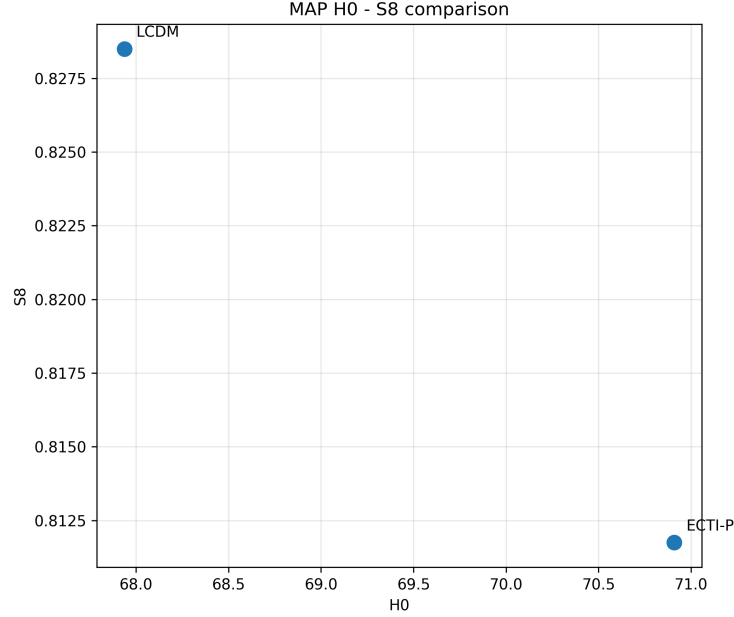


Figure 3: MAP shift in the H_0 – S_8 plane.

4.3 $\Delta\chi^2$ breakdown

The contribution of each dataset to the total $\Delta\chi^2$ is shown in Fig. 4.

At the MAP level:

$$\Delta\chi_{\text{CMB}}^2 = +0.24, \quad \Delta\chi_{\text{SN}}^2 = -15.40, \quad \Delta\chi_{\text{BAO}}^2 = +2.64, \quad \Delta\chi_{\text{RSD}}^2 = +0.34, \quad \Delta\chi_{\text{KiDS}}^2 = -3.50, \quad (5)$$

leading to:

$$\Delta\chi_{\text{total}}^2 = -15.69. \quad (6)$$

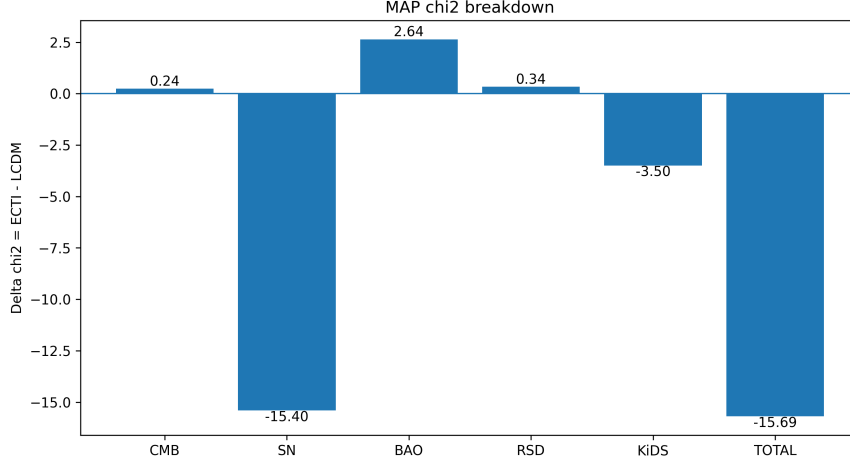


Figure 4: MAP $\Delta\chi^2$ contribution by dataset.

The improvement is dominated by Pantheon+ SN Ia, with an additional contribution from KiDS, while CMB remains nearly unchanged and BAO/RSD mildly degrade.

4.4 Tension reduction

The marginalized posterior distributions from the converged chains allow a quantitative assessment of the cosmological tensions.

For the Hubble constant, Λ CDM yields $H_0 = 67.88 \pm 0.34 \text{ km s}^{-1} \text{ Mpc}^{-1}$, in tension with the TRGB measurement of $69.8 \pm 1.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [12] at 1.11σ , and with SH0ES [13] at 4.71σ . ECTI-P shifts this to $H_0 = 70.85 \pm 0.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$, reducing the tension with TRGB to 0.60σ and with SH0ES to 1.99σ .

For the clustering amplitude, Λ CDM yields $S_8 = 0.825 \pm 0.009$, in tension with KiDS-1000 [7] at 2.58σ . ECTI-P reduces this to $S_8 = 0.809 \pm 0.009$, lowering the tension to 1.95σ .

These reductions are achieved without introducing additional free parameters in the likelihood exploration.

4.5 Expansion history

The modification of the expansion rate is shown in Fig. 5.

ECTI-P produces a localized suppression of the expansion rate at low redshift, reaching approximately -3.5% at $z = 0$.

The corresponding deformation in the dark-energy density is shown in Fig. 6.

The deviation is confined to low redshift and vanishes at higher redshift.

4.6 Distance-modulus shift

The induced shift in the predicted supernova distance modulus is shown in Fig. 7.

The model produces a negative shift reaching approximately -0.06 mag around $z \sim 0.4$.

The impact on the Pantheon+ dataset is shown directly in Fig. 8.

ECTI-P reduces structured residuals present in Λ CDM, explaining the dominant SN contribution to $\Delta\chi^2$.

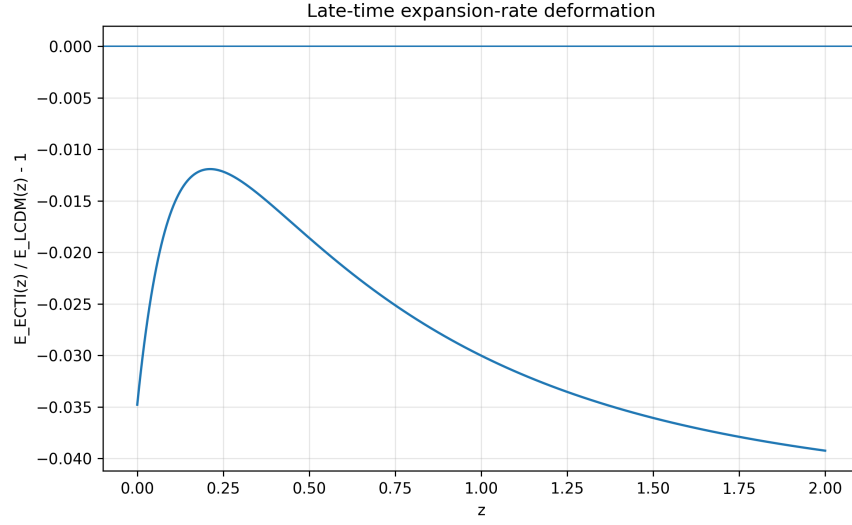


Figure 5: Relative expansion-rate deformation $E_{\text{ECTI}}(z)/E_{\text{LCDM}}(z) - 1$.

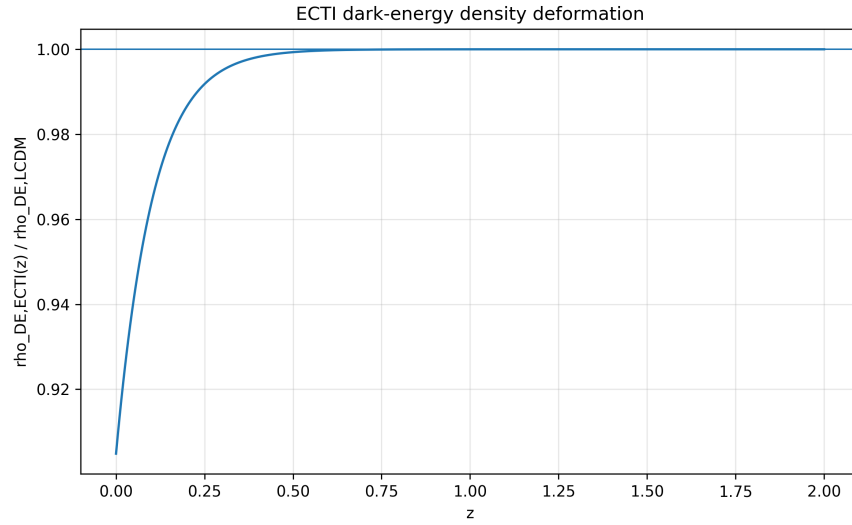


Figure 6: Relative dark-energy density $\rho_{\text{DE}}^{\text{ECTI}}(z)/\rho_{\text{DE}}^{\text{LCDM}}(z)$.

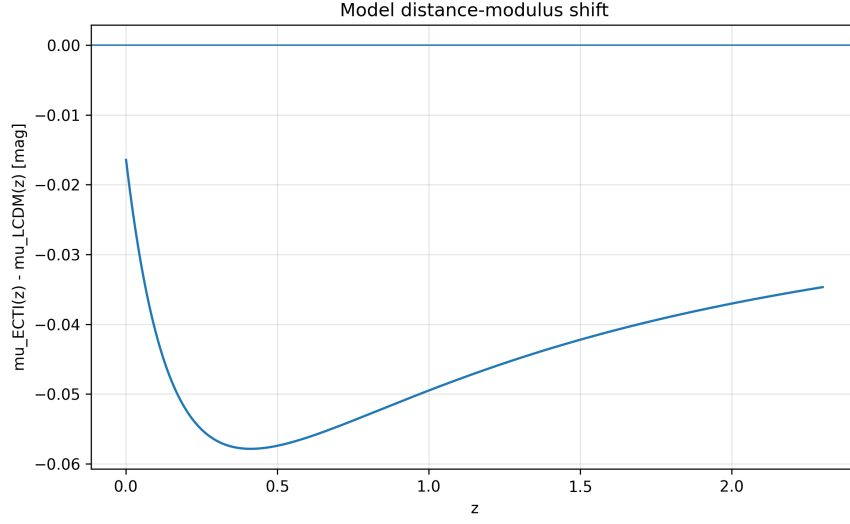


Figure 7: Distance-modulus shift $\mu_{\text{ECTI}} - \mu_{\Lambda\text{CDM}}$.

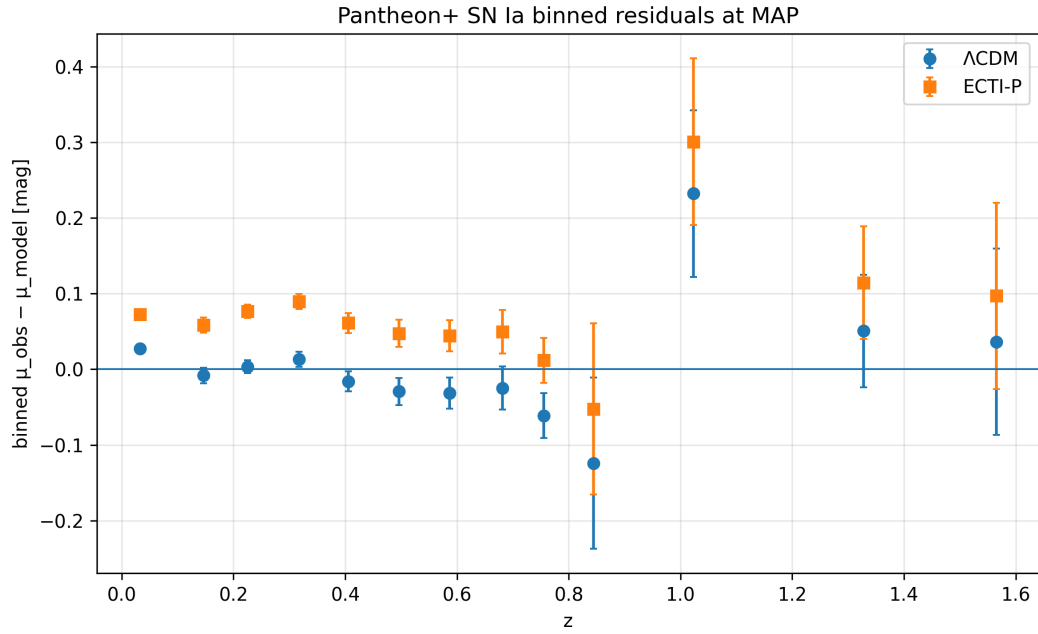


Figure 8: Binned Pantheon+ SN Ia residuals at MAP for ΛCDM and ECTI-P.

4.7 Foreground and prior diagnostic

The behavior of Planck nuisance and prior contributions at MAP is shown in Fig. 9.

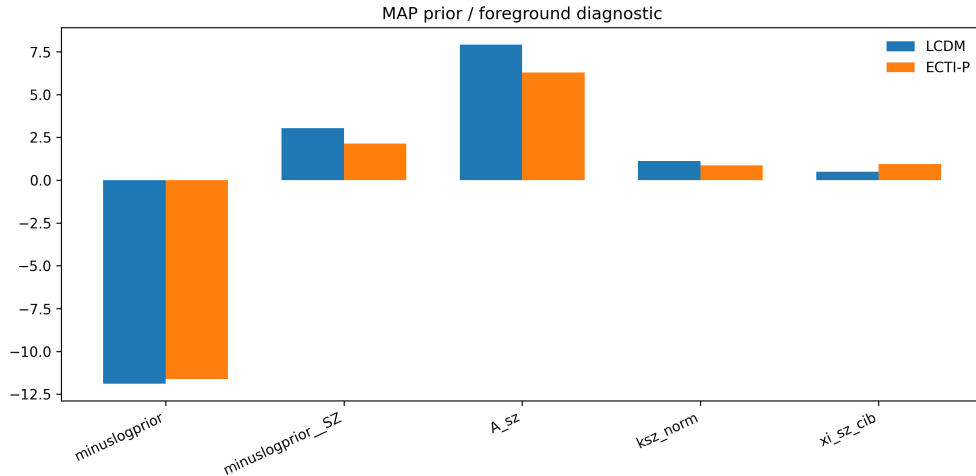


Figure 9: MAP prior and foreground parameter diagnostic.

No anomalous behavior is observed, confirming that the improvement is not driven by nuisance parameters.

4.8 Falsifiability

Future low-redshift surveys with improved SN Ia calibration and local expansion-rate measurements could directly test the predicted suppression of the expansion rate at $z \lesssim 0.2$.

The model predicts a localized deformation of $H(z)$, providing a falsifiable signature distinguishable from standard Λ CDM.

5 Discussion

The observed improvement is consistent with the physical motivation of ECTI-P: a localized suppression of the expansion rate at low redshift preferentially affects distance-sensitive probes such as Type Ia supernovae, while leaving early-Universe observables largely unaffected. The dominant contribution from Pantheon+ ($\Delta\chi^2 = -15.40$) reflects the sensitivity of SN Ia to the integrated expansion history at $z \lesssim 1$.

Several late-time dark energy extensions have been proposed to address the H_0 and S_8 tensions, including CPL parametrizations [10, 11], interacting dark energy models, and early dark energy scenarios. ECTI-P differs from these in a key respect: the deformation parameters (β, z_t) are fixed from an independent background-only calibration and are not varied in the present likelihood exploration. This reduces the effective parameter count relative to Λ CDM to zero, making the comparison strictly controlled under the Occam's razor criterion.

The moderate BAO degradation ($\Delta\chi^2_{\text{BAO}} = +2.64$) is consistent with a known tension between DESI 2024 measurements and late-time models that prefer a lower Ω_m . This represents a genuine limitation of the current model and should be investigated further, particularly in light of upcoming DESI data releases.

The hybrid perturbation treatment adopted here — standard Λ CDM perturbations on an ECTI-P background — introduces a known approximation. Dark energy perturbations are not modeled self-consistently, and the impact of this choice on observables sensitive to dark energy clustering, such as the weak lensing signal at large angular scales, cannot be fully assessed within this framework. Developing a self-consistent perturbation sector for ECTI-P remains the primary target for future theoretical work.

Finally, we note that the MAP solution favors $H_0 = 70.91 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which lies between the Planck Λ CDM value [2] and the SH0ES measurement [13], and is consistent with TRGB-based estimates [12]. Whether this shift reflects a genuine late-time deviation from Λ CDM or an artifact of the fixed parametrization requires independent confirmation with future data.

Data and Code Availability

The full analysis pipeline, including CLASS modifications, Cobaya configurations, and MCMC chains, is publicly available at:

- GitHub repository: https://github.com/f-patellis/ECTI-P_CLASS_fullprobe_vs_LCDM
- Zenodo archive (DOI): <https://zenodo.org/records/19985329>

The Zenodo archive provides a frozen, citable version of the code and results used in this work.

6 Conclusion

We have presented a full-likelihood test of the ECTI-P model, a minimal late-time deformation of Λ CDM with no additional free parameters. Under a strictly controlled FAIR comparison, ECTI-P yields $\Delta\chi^2 = -15.69$, shifts H_0 toward $70.85 \pm 0.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$ consistent with TRGB, and reduces the S_8 tension from 2.58σ to 1.95σ . ECTI-P is not proposed as a fundamental theory, but as a phenomenological framework to isolate potential late-time deviations from Λ CDM.

References

- [1] Weinberg, D. H., Mortonson, M. J., Eisenstein, D. J., et al., *Observational Probes of Cosmic Acceleration*, Phys. Rep. **530**, 87 (2013)
- [2] Planck Collaboration VI, *Planck 2018 results. VI. Cosmological parameters*, A&A **641**, A6 (2020)
- [3] Brout, D., Scolnic, D., Popovic, B., et al., *The Pantheon+ Analysis: Cosmological Constraints*, ApJ **938**, 110 (2022)
- [4] DESI Collaboration, *DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations*, arXiv:2404.03002 (2024)
- [5] Alam, S., Ata, M., Bailey, S., et al., *The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey*, MNRAS **470**, 2617 (2017)
- [6] Alam, S., Aubert, M., Avila, S., et al., *Completed SDSS-IV extended Baryon Oscillation Spectroscopic Survey: Cosmological implications from two decades of spectroscopic surveys*, Phys. Rev. D **103**, 083533 (2021)

- [7] Asgari, M., Lin, C.-A., Joachimi, B., et al., *KiDS-1000 cosmology: Cosmic shear constraints and comparison between two point statistics*, A&A **645**, A104 (2021)
- [8] Blas, D., Lesgourgues, J., & Tram, T., *The Cosmic Linear Anisotropy Solving System (CLASS) II*, JCAP **07**, 034 (2011)
- [9] Torrado, J., & Lewis, A., *Cobaya: Code for Bayesian Analysis of physical models*, JCAP **05**, 057 (2021)
- [10] Chevallier, M., & Polarski, D., *Accelerating Universes with Scaling Dark Matter*, Int. J. Mod. Phys. D **10**, 213 (2001)
- [11] Linder, E. V., *Exploring the Expansion History of the Universe*, Phys. Rev. Lett. **90**, 091301 (2003)
- [12] Freedman, W. L., Madore, B. F., Hatt, D., et al., *The Carnegie-Chicago Hubble Program. VIII. An Independent Determination of the Hubble Constant Based on the Tip of the Red Giant Branch*, ApJ **891**, 57 (2020)
- [13] Riess, A. G., Yuan, W., Macri, L. M., et al., *A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team*, ApJ Lett. **934**, L7 (2022)